

WRAPPER CLASS

Characteristics of Wrapper class:

1. It is inbuilt class, present in java.lang package.
2. It contains static and non-static methods.
3. There are 8 wrapper classes, one for each primitive datatype like Byte, Short, Integer, Long, Float, Double, Boolean and Character.
4. Wrapper classes are used to achieve 100% OOPS, that is even for these classes objects can be created.
5. For example, Byte object = new Byte ();
6. They are final classes, which means they cannot be inherited.

BOXING:

- ❖ The process of converting primitive datatype value into an object reference of its corresponding Wrapper class type is called Boxing.
- ❖ A static method called `valueOf(primitive_value)` is used.

```
byte b = 12;  
  
Byte obj = Byte.valueOf(b);
```

UNBOXING:

- ❖ The process of converting the Wrapper class type object reference back to its corresponding primitive datatype value is called Unboxing. Boxing is a pre-condition for Unboxing.
- ❖ A non-static method called `value()` is used.

```
Byte obj = Byte.valueOf(b);  
  
byte b = obj.value();
```

PROGRAM EXAMPLE FOR BOXING AND UNBOXING:

```
1 package collection.interfaces;  
2  
3 public class SampleDemo {  
4     public static void main(String[] args) {  
5         byte b = 12;  
6         Byte obj = Byte.valueOf(b);  
7         System.out.println(obj);  
8         byte b1 = obj.byteValue();  
9         System.out.println(b1);  
10        short s = obj.shortValue();  
11        System.out.println(s);  
12    }  
13 }  
14 }
```

```
<termini  
12  
12  
12  
12
```